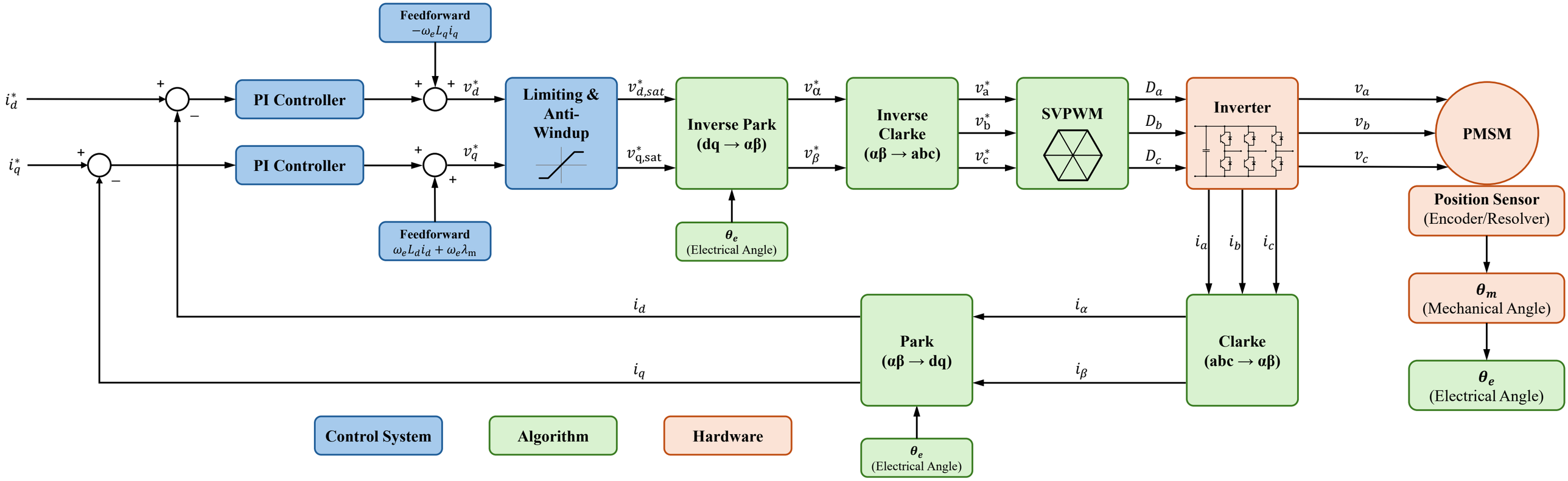

Feedforward
$-\omega_e L_q i_q$
i_d^*
i_q^*
PI Controller
PI Controller
v_d^*
v_q^*
Feedforward
$\omega_e L_d i_d + \omega_e \lambda_m$
Limiting & Anti-Windup
$v_{d,sat}^*$
$v_{q,sat}^*$
Inverse Park (dq → αβ)
θ_e (Electrical Angle)
v_α^*
v_β^*
Inverse Clarke (αβ → abc)
v_a^*
v_b^*
v_c^*
SVPWM
D_a
D_b
D_c
Inverter
v_a
v_b
v_c
PMSM
Position Sensor (Encoder/Resolver)
θ_m (Mechanical Angle)
θ_e (Electrical Angle)
i_a
i_b
i_c
Clarke (abc → αβ)
i_α
i_β
Park (αβ → dq)
i_d
i_q
θ_e (Electrical Angle)
Control System
Algorithm
Hardware